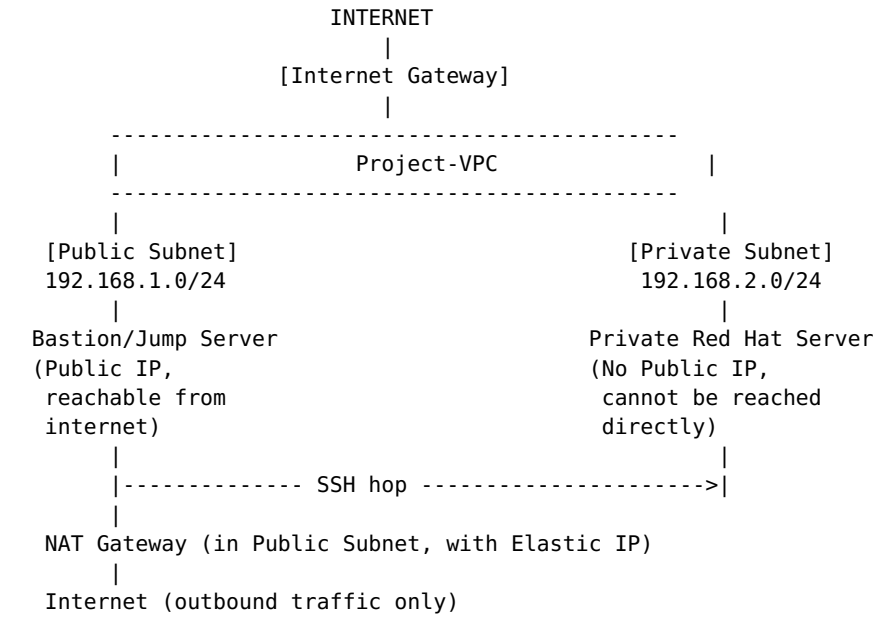
Custom VPC with Bastion Host + NAT Gateway (Secure Network Architecture)

Project Overview

This project builds a secure, segmented AWS network from scratch — a custom VPC with public and private subnets, where a **private Red Hat EC2 instance** has no direct internet exposure and can only be reached through a **Bastion Host**, while still being able to reach the internet outbound (for updates) through a **NAT Gateway**.

This mirrors how real companies isolate sensitive resources (databases, internal apps) from the public internet while still keeping them patchable and manageable.

Architecture Diagram



Key AWS Services Used

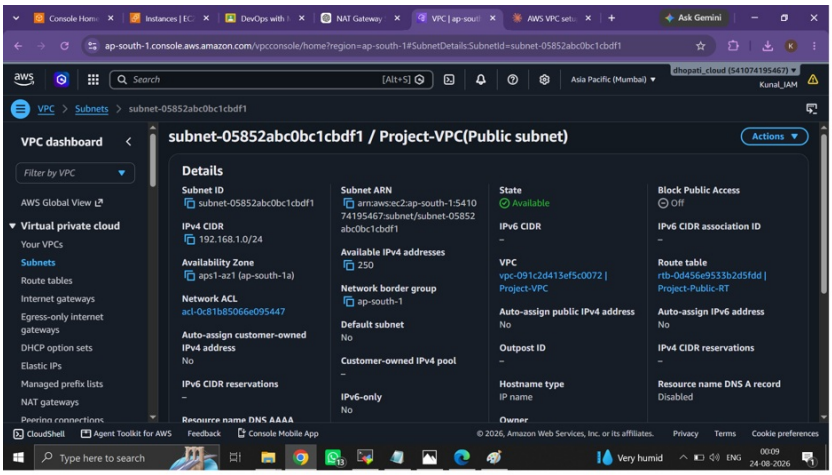
Service	Purpose
Amazon VPC	Custom isolated network (`Project-VPC`)
Public & Private Subnets	Segregates internet-facing vs internal resources
Internet Gateway (IGW)	Allows public subnet to reach/be reached from the internet
NAT Gateway + Elastic IP	Gives the private subnet outbound-only internet access
Bastion Host	Single controlled entry point (jump server) into the private network
	Controls SSH access at the

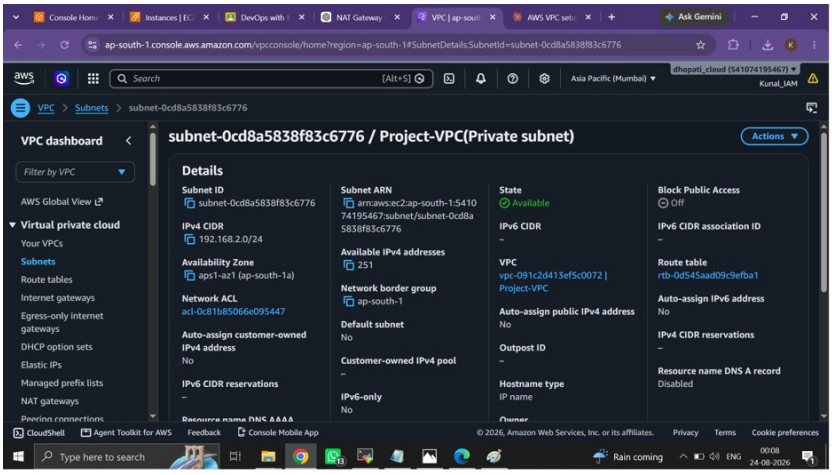
Security Groups	network level
Amazon EC2 (Red Hat)	Private application server, patched via `dnf update`

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1. VPC & Subnets

- **VPC:** `Project-VPC`
- **Public Subnet:** `192.168.1.0/24` (ap-south-1a) — hosts the Bastion Host
- **Private Subnet:** `192.168.2.0/24` (ap-south-1a) — hosts the Red Hat application server

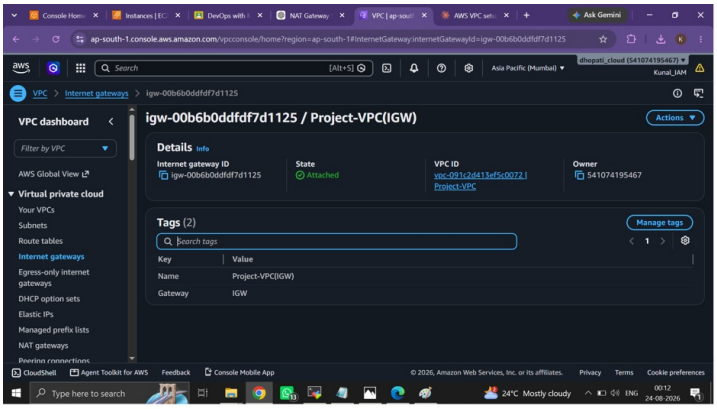




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2. Internet Gateway

`Project-VPC(IGW)` — Attached to `Project-VPC`, giving the public subnet a path to and from the internet.

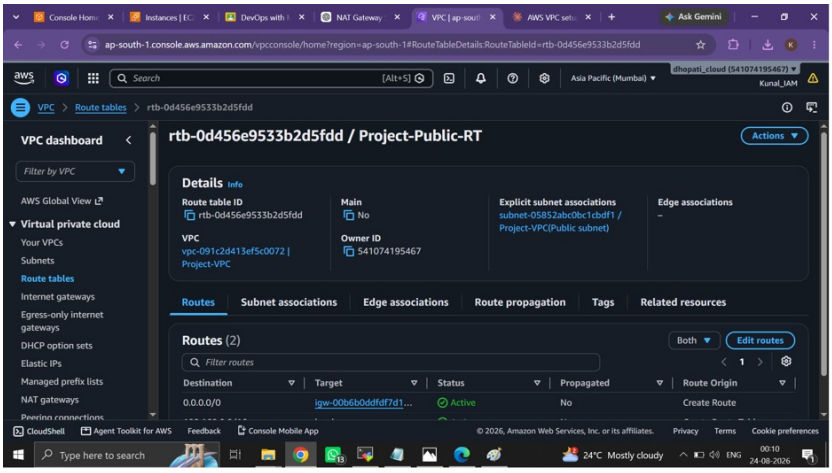


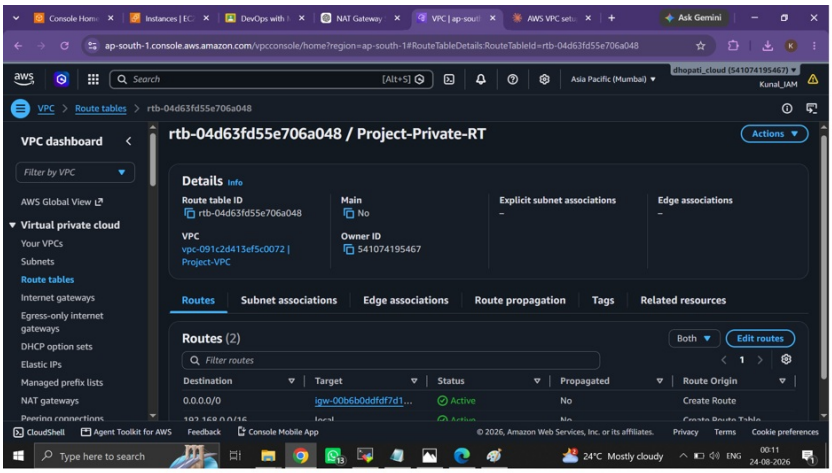
Internet Gateway — Attached

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3. Route Tables

- **Project-Public-RT** — routes `0.0.0.0/0` → Internet Gateway (used by the public subnet / Bastion Host)
- **Project-Private-RT** — associated with the private subnet, controlling how the Red Hat server's traffic is routed for outbound access

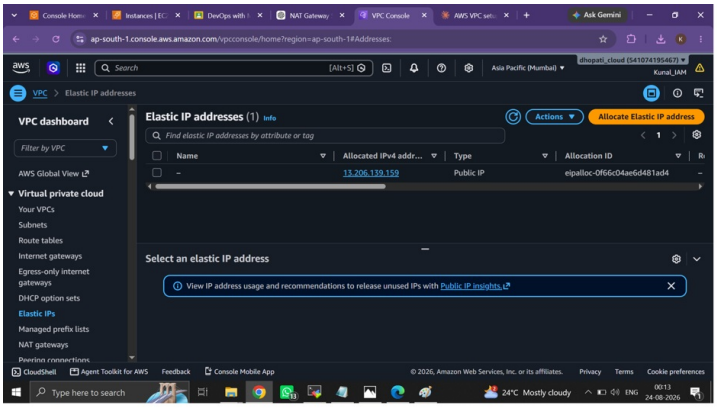




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4. NAT Gateway + Elastic IP

A NAT Gateway was placed in the public subnet with a dedicated Elastic IP (`13.206.139.159`). This allows the private Red Hat server to reach the internet for outbound requests (like package updates) **without** ever being directly reachable from the internet — traffic only flows outbound, never inbound.

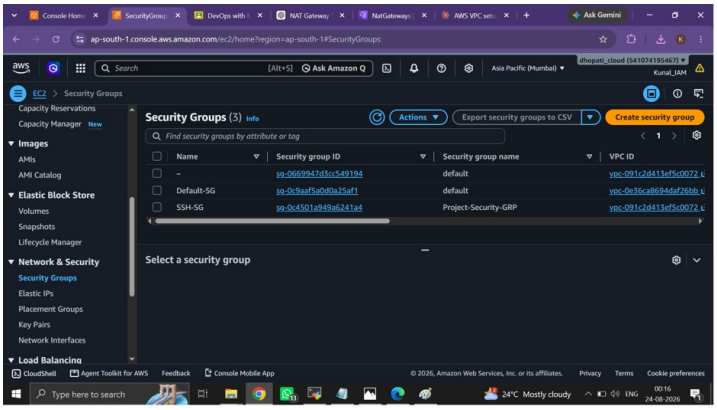


NAT Gateway Elastic IP

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5. Security Groups

`SSH-SG` (Project-Security-GRP) — controls SSH access to the Bastion Host, alongside default VPC security groups.



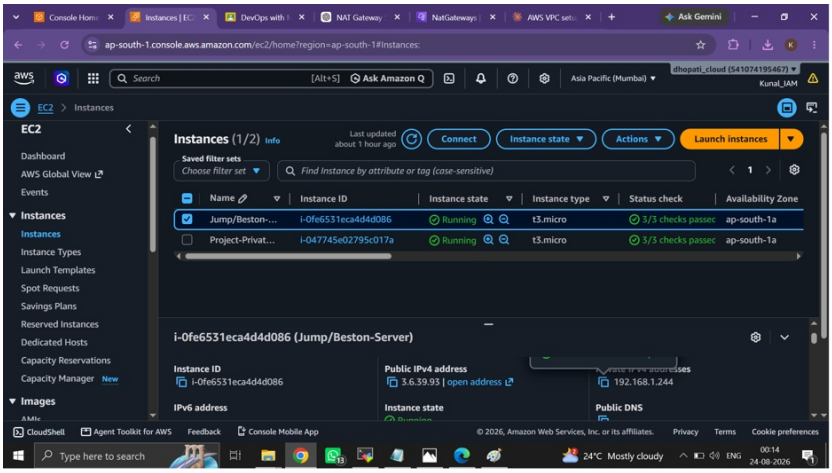
Security Groups

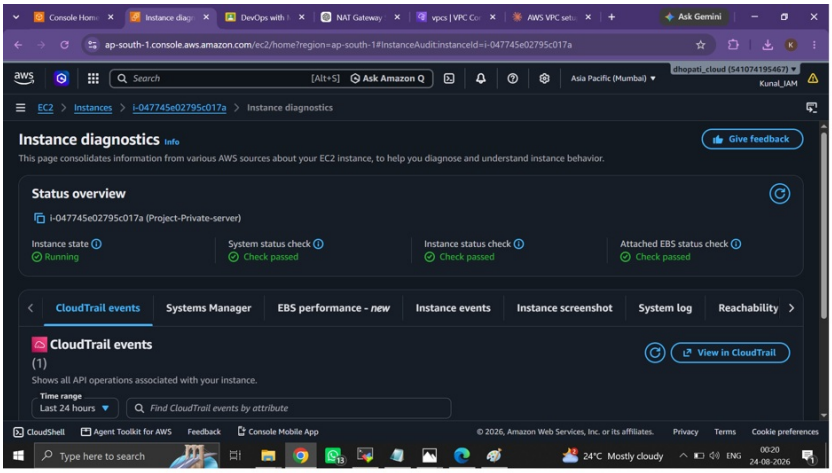
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6. EC2 Instances

Two instances launched: - **Jump/Beston-Server** — in the public subnet, has a public IP, reachable directly via SSH - **Project-Private-server** — in the private subnet, no public IP, reachable only via the Bastion Host

Both instances show **Running** state with all status checks passed.





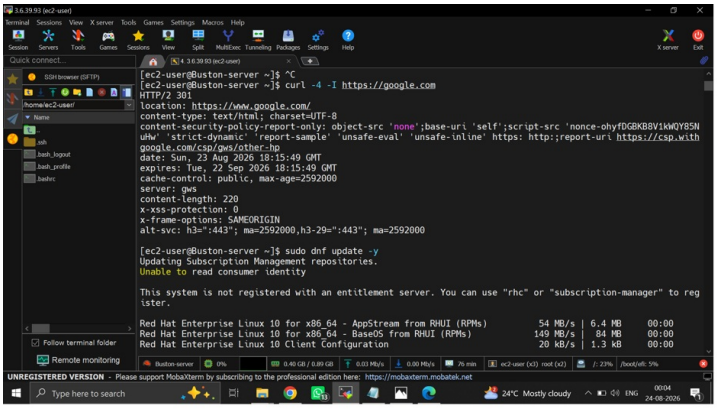
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7. Verification — Bastion Connectivity & NAT Gateway Test

From the Bastion server terminal:

```
curl -4 -I https://google.com      # confirms outbound internet access
sudo dnf update -y                 # confirms package manager can reach Red Hat repos via NAT
```

Both commands succeeded, confirming the Bastion Host has internet access and the private server (routed through NAT Gateway) can successfully reach external repositories for updates.



Bastion Terminal — Connectivity Test

How It Works — Simple Explanation

1. VPC = Your own house/office. A big network boundary was created — everything inside it is isolated from other AWS networks.

2. Public Subnet vs Private Subnet. Public Subnet = the reception area — reachable directly from outside. Private Subnet = the inner room — nobody from outside can enter directly.

3. Internet Gateway (IGW) = The main door. Connects the VPC to the internet; only the public subnet uses this door.

4. NAT Gateway + Elastic IP = The courier. The private server can’t go outside on its own, so the NAT Gateway (sitting in the public subnet) fetches things (like updates) on its behalf and delivers them back. This is **outbound-only** — nobody from outside can come in through it.

5. Bastion Host = The security guard’s cabin. The only server with a public IP; admins SSH into it directly, then jump from there to the private server.

6. Private Red Hat EC2 = The inner room’s server. No public IP, cannot be touched directly from the internet, reachable only through the Bastion.

7. Admin → Bastion → Private Server is the access path — you can never reach the private server directly from outside.

8. DNS Resolution worked correctly inside the VPC — internal name resolution was functioning.

9. Private Server → NAT Gateway → Internet: when `dnf update` ran on the private server, it asked the NAT Gateway to fetch updates, which it did and delivered back — confirming one-way internet access was working correctly.

Why This Setup Matters (Real-World Use)

In real companies, sensitive resources like database servers and internal applications are never placed directly on the internet. They’re kept private, just like this, and access is given only through a controlled gate (Bastion). This project demonstrates that exact pattern hands-on.

One-Line Summary (For Interviews)

> "I built a VPC with public and private subnets. The Red Hat server in the private subnet had no direct access from outside — it could only be reached via SSH through the Bastion Host. The private server needed internet access for updates, so I used a NAT Gateway, which allows only outbound traffic, not inbound."

Skills Demonstrated

`VPC` · `Public/Private Subnet` · `Internet Gateway` · `NAT Gateway` · `Bastion Host` · `EC2` · `Security Groups` · `SSH` · `DNS`

Project Status

End-to-end functional and verified — private instance is fully isolated from direct internet access, reachable only via Bastion, with confirmed outbound-only internet connectivity through the NAT Gateway (validated via a live `dnf update`).

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